

Week 12

Deep Learning

Question: 1

A translation system without attention translates the French sentence "*Les étudiants travaillent dans la bibliothèque*" into English. However, it often produces incomplete or grammatically incorrect translations for long sentences.

Which issue in the seq2seq model does this most likely illustrate?

Options:

- A) Limited size of vocabulary
- B) Overfitting to training data
- C) Loss of important information due to fixed-length context vector
- D) Incorrect tokenization method

Correct Answer: C

Question: 2

Which of the following are valid motivations for introducing attention in seq2seq models?

Options:

- A) To allow the decoder to dynamically select relevant encoder outputs for each output timestep
- B) To completely remove the need for an encoder
- C) To overcome the bottleneck of fixed-length context vectors
- D) To ensure that the model ignores irrelevant tokens in the input sequence

Correct Answer: A, C, D

Question: 3

An English-to-German translation model is observed to generate the German equivalent of "He ate the cake" as "Er aß den Kuchen" with correct word order. Attention weight visualization shows high weights between "cake" and "Kuchen," and between "ate" and "aß." What does this indicate about the role of attention?

Options:

- A) Attention ensures perfect translation without any need for training
- B) Attention aligns relevant source and target tokens during decoding
- C) Attention forces the model to translate words in exactly the same order as the source language
- D) Attention is only useful for short sentences

Correct Answer: B

Question: 4

In Bahdanau attention for NMT, how is the alignment score between the encoder hidden state h_i and decoder hidden state s_t typically computed?

Options:

- A) Using a fixed lookup table
- B) By a feedforward network with trainable parameters
- C) By simple addition of h_i and s_t
- D) By computing cosine similarity without learning parameters

Correct Answer: B

Question: 5

A sequence-to-sequence model is generating a weather report from satellite data. At decoding step t , the model calculates scores between the decoder's current hidden state and each encoder hidden state, then applies a softmax to obtain weights.

What is the purpose of these weights?

Options:

- A) To decide the vocabulary size for output generation
- B) To determine how much each encoder hidden state contributes to the context vector
- C) To initialize the decoder's hidden state for the next time step
- D) To normalize the encoder outputs to have zero mean

Correct Answer: B

Question: 6

What happens after the attention weights are computed in additive attention?

Options:

- A) They are multiplied by each encoder hidden state and summed to form the context vector
- B) They replace the decoder hidden state entirely
- C) They are concatenated to the input sequence
- D) They are stored for backpropagation but not used in the current timestep

Correct Answer: A

Question: 7

A news headline generator model is using additive attention. At a certain decoding step, the model assigns nearly equal attention weights to all encoder hidden states.

Which of the following is the most likely reason?

Options:

- A) All input tokens are equally relevant for generating the next word
- B) The model has failed to learn the attention parameters
- C) The encoder has produced identical hidden states for all tokens
- D) The vocabulary is too large

Correct Answer: A

Question: 8

In additive attention, why is a softmax used after computing alignment scores?

Options:

- A) To normalize the scores into probabilities that sum to 1
- B) To make the attention process faster
- C) To prevent the decoder from using the same encoder state repeatedly
- D) To reduce the number of encoder hidden states

Correct Answer: A

Question: 9

A machine translation model uses a scoring function that calculates the dot product between the decoder hidden state s_t and each encoder hidden state h_i to determine alignment. What is the main advantage of this scoring method?

Options:

- A) It introduces additional trainable parameters for better alignment
- B) It is computationally faster because it avoids extra parameters
- C) It eliminates the need for normalization
- D) It forces equal attention weights for all positions

Correct Answer: B

Question: 10

Which of the following are valid variants of multiplicative attention scoring?

Options:

- A) Dot product between s_t and h_i
- B) Weighted dot product using a trainable matrix

- C) Addition of s_t and h_i followed by a feedforward layer
- D) Cosine similarity without any learning parameters

Correct Answer: A, B

Question: 11

A translation model engineer switches from additive to multiplicative attention and observes a speedup in training without a drop in accuracy.

Which is the most likely reason for the improvement?

Options:

- A) Multiplicative attention ignores most encoder outputs
- B) Multiplicative attention uses fewer or no extra parameters, reducing computation
- C) Multiplicative attention replaces the decoder with a CNN
- D) Multiplicative attention forces the model to output shorter sentences

Correct Answer: B

Question: 12

In the “general” form of multiplicative attention, how is the score between s_t and h_i computed?

Options:

- A) $s_t^T W h_i$, where W is a trainable weight matrix
- B) $s_t^T h_i$ without parameters
- C) Addition of s_t and h_i followed by tanh activation
- D) Multiplication of s_t by a scalar

Correct Answer: A

Question: 13

A researcher visualizes the attention weights of a translation model as a heatmap. The heatmap shows a strong diagonal pattern from the top-left to the bottom-right. What does this pattern indicate?

Options:

- A) The model is ignoring all source words
- B) The model is translating word-by-word in approximately the same order
- C) The model is overfitting to the training set
- D) The model has a vanishing gradient problem

Correct Answer: B

Question: 14

Why are attention weights often visualized as heatmaps?

Options:

- A) To directly debug gradient values
- B) To visually inspect which source tokens the decoder focuses on during each output step
- C) To replace numerical evaluation metrics
- D) To determine the vocabulary size of the model

Correct Answer: B

Question: 15

Which of the following statements are true about attention weight visualizations?

Options:

- A) They can reveal if the model is focusing on irrelevant tokens
- B) They can help identify whether the model has learned meaningful alignments
- C) They are generated before the softmax step
- D) They can be used to interpret model behavior qualitatively

Correct Answer: A, B, D

Question: 16

An attention heatmap for a Japanese-to-English model shows scattered high-weight areas rather than a clear diagonal or block pattern.

Which of the following could be a reasonable explanation?

Options:

- A) The source and target languages have very similar word order
- B) The translation involves significant word reordering due to language structure differences
- C) The model is using a fixed-length context vector without attention
- D) The softmax function is not applied

Correct Answer: B

Question: 17

A company is building a document classification model that must capture relationships between words regardless of their distance in the sentence. The team decides to use self-attention instead of an RNN-based encoder.

Which property of self-attention directly supports this decision?

Options:

- A) It processes words sequentially to preserve time-step order
- B) It models dependencies between all tokens in parallel, regardless of distance

- C) It reduces the vocabulary size automatically
- D) It forces all tokens to have equal attention weights

Correct Answer: B

Question: 18

In self-attention, what are the roles of the Query (Q), Key (K), and Value (V) matrices?

Options:

- A) Query and Key compute attention scores; Values are combined using the weights
- B) Query stores the output vocabulary; Key stores the input vocabulary; Value stores translations
- C) All three are identical copies of the input embeddings
- D) Query and Value compute attention scores; Keys are ignored

Correct Answer: A

Question: 19

Given a query vector $Q = [1, 2]$ and a key vector $K = [2, 1]$.

If the dimensionality $d_k = 4$, what is the scaled dot product score before softmax?

Options:

- A) 4
- B) 2
- C) 1
- D) 8

Correct Answer: B

Question: 20

An NLP engineer visualizes self-attention weights in a transformer model and notices that the word “bank” attends strongly to both “river” and “shore” in the same sentence.

What does this tell us about self-attention?

Options:

- A) It can represent multiple contextual meanings of a word simultaneously
- B) It always forces words to attend to themselves
- C) It ignores polysemy in language
- D) It computes attention only for adjacent tokens

Correct Answer: A

Question: 21

Why is the scaling factor $\frac{1}{\sqrt{d_k}}$ used in scaled dot-product attention?

Options:

- A) To increase the magnitude of the scores for better gradients
- B) To prevent large dot product values from pushing softmax into regions with very small gradients
- C) To normalize the sequence length
- D) To make all attention weights equal

Correct Answer: B

Question: 22

A translation team moves from an RNN-based encoder-decoder to a transformer-based model. They immediately notice much faster training times on GPUs. Which feature of the transformer architecture is the primary reason for this speedup?

Options:

- A) Removal of the decoder network
- B) Processing of sequence tokens in parallel using self-attention
- C) Smaller vocabulary size in transformers
- D) Pre-training on fewer examples

Correct Answer: B

Question: 23

Why is multi-head attention used in transformer architectures?

Options:

- A) To increase model size without improving performance
- B) To allow the model to focus on different representation subspaces at different positions
- C) To eliminate the need for feedforward layers
- D) To speed up matrix multiplication

Correct Answer: B

Question: 24

Which of the following statements are true about **positional encoding** in transformers?

Options:

- A) It adds position information to token embeddings
- B) It can be learned or fixed (e.g., sinusoidal)

- C) It replaces the need for embeddings entirely
- D) It ensures the model can differentiate between tokens in different positions

Correct Answer: A, B, D

Question: 25

A transformer uses 8 attention heads, each with a key/query dimension of 64.
What is the total concatenated dimension before passing through the output projection layer?

Options:

- A) 64
- B) 128
- C) 256
- D) 512

Correct Answer: D

Question: 26

A speech recognition model is struggling to transcribe long audio sequences accurately. An engineer proposes adding an attention mechanism.
Which is the most likely benefit?

Options:

- A) The model can ignore all past inputs
- B) The model can focus on the most relevant audio frames for each output token
- C) The model can eliminate the need for acoustic features
- D) The model will reduce its vocabulary size

Correct Answer: B

Question: 27

Which of the following are real-world applications where attention mechanisms are widely used?

Options:

- A) Neural Machine Translation (NMT)
- B) Image Captioning
- C) Speech Recognition
- D) Sorting numerical arrays

Correct Answer: A, B, C

Question: 28

In an image captioning model, the attention heatmap shows high weights over the "ball" region in the image while generating the word "ball" in the caption.
What does this indicate?

Options:

- A) The attention mechanism failed to learn spatial focus
- B) The model is attending to irrelevant parts of the image
- C) The model is correctly focusing on the visual region relevant to the word being generated
- D) The attention weights are being computed incorrectly

Correct Answer: C

Question: 29

Why is attention particularly beneficial in Neural Machine Translation for long sentences?

Options:

- A) It reduces the training data requirements
- B) It removes the encoder entirely
- C) It allows the decoder to dynamically focus on relevant parts of the source sentence at each step
- D) It ensures translations always have the same length as the source sentence

Correct Answer: C

Question: 30

A medical image diagnosis model uses attention to highlight regions of an X-ray that influenced its prediction.

Which additional benefit does this provide besides improving accuracy?

Options:

- A) Increased training speed
- B) Model interpretability for human experts
- C) Reduced image resolution requirements
- D) Automatic labeling of training data

Correct Answer: B